RETAIL SALES DATA ANALYSIS

Technical Project Report

Internship: Data Analytics in Python

Organization: Sumerix Global

Project Title: Retail Sales Data Exploration & Analysis Dashboard

Tools Used: Python, Pandas, NumPy, Matplotlib, Power BI, Microsoft Excel

Submitted By: Shweta Mohan Khandagale

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Project Type: Data Analytics Internship Project

1. PROJECT OVERVIEW

Retail businesses generate large amounts of sales data every day. Analyzing this data helps businesses understand customer purchasing behavior, product performance, revenue trends, and payment preferences.

This project focuses on exploring and analyzing retail sales data using Python and Power BI.

The main purpose of this project is to transform raw retail transaction data into meaningful business insights through data cleaning, exploratory data analysis, data visualization, and dashboard development.

Python libraries such as Pandas, NumPy, and Matplotlib were used for data processing and analysis. Power BI was used to create an interactive dashboard to present important sales information.

The project helps identify high-performing product categories, popular items, customer activity, payment preferences, and sales trends.

2. PROBLEM STATEMENT

Retail businesses need to understand their sales performance to make better business decisions. However, raw transaction data alone does not clearly explain which products generate the most revenue, which payment methods are preferred, or how sales change over time.

The main problem addressed in this project is the difficulty of extracting meaningful information from raw retail sales data.

This project uses Python and Power BI to analyze the dataset and answer important business questions.

Key Questions:

1. What is the total revenue generated?
2. Which product categories generate the highest revenue?
3. Which payment method is used most frequently?
4. Which sales channel performs better: Online or In-store?
5. Which items have the highest sales quantity?

6. How does revenue change monthly and yearly?
7. How can interactive dashboards help understand sales performance?

The analysis provides useful insights that can support business planning and decision-making.

3. PROJECT OBJECTIVES

The main objectives of this Retail Sales Data Analysis project are:

1. **Data Collection:** Understand and load the retail sales dataset into Python.
2. **Data Cleaning:** Identify and handle missing values, incorrect data types, and duplicate records.
3. **Data Analysis:** Calculate total revenue, average transaction value, transaction count, and customer count.
4. **Category Analysis:** Identify the product categories generating the highest revenue.
5. **Payment Analysis:** Understand customer payment preferences.
6. **Location Analysis:** Compare sales performance between Online and In-store channels.
7. **Trend Analysis:** Analyze monthly and yearly revenue patterns.
8. **Data Visualization:** Create meaningful charts using Python.
9. **Dashboard Development:** Build an interactive Power BI dashboard with KPI cards, charts, and slicers.
10. **Business Insights:** Present findings that can support better business decisions.

4. DATASET INFORMATION

The dataset used in this project contains retail store transaction records. Each row represents a sales transaction and provides information about the customer, product, quantity, payment method, and transaction date.

Dataset Details:

- Original dataset name: retail_store_sales.csv
- Original number of records: 12,575

- Number of columns: 11
- Records after data cleaning: 11,362
- Final number of columns: 11
- Missing values after cleaning: 0
- Duplicate rows after cleaning: 0

Important Dataset Columns:

Column	Description
Transaction ID	Unique transaction identifier
Customer ID	Customer identifier
Category	Product category
Item	Product name
Price Per Unit	Price of one unit
Quantity	Number of units purchased
Total Spent	Total transaction amount
Payment Method	Payment method used
Location	Online or In-store
Transaction Date	Date of transaction
Discount Applied	Discount information

The cleaned dataset was saved as a CSV file and used for further analysis and Power BI dashboard development.

5. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used to complete this project.

1. Python

Python was used as the main programming language for data cleaning, analysis, and visualization.

2. Pandas

Pandas was used to load CSV files, handle missing values, remove unnecessary records, and perform data analysis.

3. NumPy

NumPy was used for numerical operations and calculations.

4. Matplotlib

Matplotlib was used to create charts and graphs to understand sales performance.

5. Jupyter Notebook

Jupyter Notebook was used to write and execute Python code and document the analysis process.

6. Microsoft Excel

Excel was used to organize and present the analysis results in multiple worksheets.

7. Power BI

Power BI was used to create an interactive dashboard containing KPI cards, charts, and slicers for exploring retail sales performance.

6. DATA CLEANING AND PREPROCESSING

Data cleaning is an important step in data analysis. The original retail sales dataset contained 12,575 records. Before performing analysis, the data was checked and cleaned using Python and Pandas.

Cleaning Steps Performed:

1. Loading the Dataset

The original CSV file was loaded into Python using the Pandas library.

2. Handling Missing Values

Missing values in the dataset were checked. Missing values in the Discount Applied and Item columns were filled with "Unknown".

3. Handling Numerical Missing Values

Rows containing missing values in Price Per Unit, Quantity, and Total Spent were removed.

4. Date Conversion

The Transaction Date column was converted into a proper datetime format for time-based analysis.

5. Duplicate Checking

Duplicate records were checked. No duplicate rows were found in the cleaned dataset.

6. Saving the Cleaned Dataset

The cleaned dataset was saved as a CSV file for further analysis and Power BI dashboard development.

Final Cleaning Results:

- Original records: 12,575
- Records after cleaning: 11,362
- Columns: 11
- Missing values: 0
- Duplicate rows: 0

The cleaning process improved the quality and reliability of the dataset for further analysis.

7. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed using Python to understand the retail sales dataset and identify important patterns.

Different analysis techniques were used to examine sales performance, customer activity, product categories, payment methods, and revenue trends.

Analysis Performed:

1. Overall Sales Analysis

Calculated total revenue, total transactions, total customers, and average transaction value.

2. Category Analysis

Compared revenue generated by different product categories to identify the highest-performing category.

3. Payment Method Analysis

Analyzed sales across Cash, Credit Card, and Digital Wallet payment methods.

4. Location Analysis

Compared Online and In-store sales to understand channel performance.

5. Item Analysis

Identified the best-selling items based on quantity sold.

6. Monthly and Yearly Analysis

Analyzed revenue patterns across different months and years.

7. Customer Analysis

Examined customer transaction activity to understand purchasing behavior.

The results of these analyses were used to create charts and identify meaningful business insights.

8. OVERALL SALES PERFORMANCE

The overall sales performance was analyzed using the cleaned retail sales dataset.

The following key performance indicators (KPIs) were calculated:

Metric	Result
Total Revenue	1,472,998.50
Total Transactions	11,362
Total Customers	25
Average Transaction Value	129.64
Maximum Transaction Value	410
Minimum Transaction Value	5

Key Observations:

1. The business generated total revenue of 1,472,998.50.
2. A total of 11,362 transactions were recorded.
3. The dataset contains 25 unique customers.
4. The average transaction value was 129.64.
5. The highest transaction value was 410, while the lowest was 5.

These results provide an overview of the overall retail sales performance.

9. SALES ANALYSIS BY PRODUCT CATEGORY

Category-wise sales analysis was performed to identify which product categories contributed the highest revenue.

Revenue by Category:

Product Category	Total Revenue
Butchers	197,426.00
Electric household essentials	192,441.50
Beverages	187,978.50
Furniture	186,527.00
Food	184,645.00
Computers and electric accessories	180,902.50
Patisserie	172,330.50
Milk Products	170,747.50

Key Observations:

1. Butchers generated the highest revenue of 197,426.00.
2. Electric household essentials ranked second.
3. Beverages also contributed significantly to total revenue.

4. Milk Products generated the lowest revenue among the listed categories.

Business Insight:

The Butchers category performed best in terms of revenue. The business can study its products and sales strategies to understand what drives its performance. Lower-revenue categories can be reviewed for possible improvement.

10. SALES ANALYSIS BY PAYMENT METHOD

Payment method analysis was performed to understand how customers paid for their purchases.

The dataset contains three payment methods: Cash, Credit Card, and Digital Wallet.

Revenue by Payment Method:

Payment Method	Total Revenue
Cash	513,676.00
Credit Card	481,135.00
Digital Wallet	478,187.50

Key Observations:

1. Cash generated the highest revenue of 513,676.00.
2. Credit Card generated 481,135.00 in revenue.
3. Digital Wallet generated 478,187.50 in revenue.
4. Cash was the leading payment method in terms of revenue.

Business Insight:

Cash payments contributed the highest revenue, while Credit Card and Digital Wallet payments also contributed significantly. The business should continue supporting all three payment methods to provide convenience to customers.

11. SALES ANALYSIS BY LOCATION

Location-wise sales analysis was performed to compare the performance of Online and In-store sales channels.

Revenue by Location:

Sales Channel	Total Revenue
Online	749,431.00
In-store	723,567.50

Revenue Contribution:

- Online: 50.88%
- In-store: 49.12%

Key Observations:

1. Online sales generated the highest revenue of 749,431.00.
2. In-store sales generated 723,567.50.
3. Online sales contributed 50.88% of total revenue.
4. Both sales channels performed relatively equally.

Business Insight:

Online sales performed slightly better than In-store sales. The business can continue improving its online shopping experience while maintaining its physical store operations.

12. BEST-SELLING ITEMS ANALYSIS

Item-wise sales analysis was performed to identify the products purchased in the highest quantities.

The analysis used the Quantity column to determine which items were the most popular among customers.

Key Findings:

- Best-selling item: Item_2_BEV
- Total quantity sold: 676 units

Key Observations:

1. Item_2_BEV recorded the highest sales quantity of 676 units.
2. The analysis helps identify products with strong customer demand.
3. Identifying popular items can help businesses plan inventory and product availability.

Business Insight:

The business should maintain sufficient stock of popular items such as Item_2_BEV to meet customer demand and reduce the possibility of stock shortages.

13. MONTHLY AND YEARLY SALES ANALYSIS

Time-based sales analysis was performed to understand revenue trends across different months and years.

Monthly Analysis:

- Highest revenue month: January
- Lowest revenue month: October

Yearly Revenue Analysis:

Year	Total Revenue
2022	484,301.50
2023	465,149.50
2024	499,336.00
2025	24,211.50

Key Observations:

1. January recorded the highest monthly revenue.
2. October recorded the lowest monthly revenue.
3. 2024 generated the highest annual revenue among the listed years.
4. Revenue in 2025 represents only January data, so it should not be compared directly with full-year totals.

Business Insight:

The business can use monthly and yearly revenue trends to identify seasonal patterns, plan inventory, and improve sales strategies. Additional data from future months will help provide a more complete yearly comparison.

14. DATA VISUALIZATION

Data visualization was performed to present the analysis results in an easy-to-understand graphical format.

Python's Matplotlib library was used to create charts and graphs during exploratory data analysis.

Visualizations Created:

1. Revenue by Product Category
2. Revenue by Payment Method
3. Revenue by Location
4. Top-Selling Items
5. Monthly Revenue Trend
6. Yearly Revenue Comparison
7. Customer and Transaction Analysis

These visualizations helped identify sales patterns, compare different categories, and understand business performance.

Charts make complex data easier to understand and help communicate important findings effectively.

15. POWER BI DASHBOARD DEVELOPMENT

An interactive Retail Sales Analysis Dashboard was developed using Microsoft Power BI.

The cleaned retail sales dataset was imported into Power BI, and appropriate data types and DAX measures were created.

Key Performance Indicators (KPIs):

1. Total Revenue

2. Total Transactions
3. Total Customers
4. Average Transaction Value

Dashboard Visualizations:

1. Revenue by Product Category
2. Revenue by Payment Method
3. Revenue by Sales Channel
4. Top 10 Best-Selling Items
5. Monthly Revenue Trend
6. Yearly Revenue Comparison

Interactive Filters (Slicers):

- Location
- Category
- Payment Method
- Transaction Date

These slicers allow users to filter the dashboard and explore sales performance based on different conditions.

The dashboard provides a clear visual summary of retail business performance and makes it easier to explore important sales information.

16. KEY BUSINESS INSIGHTS

Based on the retail sales analysis, the following important business insights were identified:

1. **Strong Overall Revenue:** The business generated total revenue of 1,472,998.50 from 11,362 transactions.
2. **Best Performing Category:** Butchers generated the highest category revenue of 197,426.00.
3. **Preferred Payment Method:** Cash generated the highest revenue of 513,676.00.

4. **Best Performing Sales Channel:** Online sales performed slightly better than In-store sales, generating 749,431.00.
5. **Popular Product:** Item_2_BEV was the best-selling item, with 676 units sold.
6. **Monthly Performance:** January recorded the highest monthly revenue, while October recorded the lowest.
7. **Yearly Performance:** 2024 recorded the highest yearly revenue of 499,336.00 among the years analyzed.

Overall Business Recommendation:

The business should maintain sufficient inventory for popular products, study the success of high-performing categories, improve its online sales channel, and monitor monthly sales trends to support better business decisions.

17. CONCLUSION

The Retail Sales Data Analysis project successfully demonstrated how data analytics tools can be used to understand retail business performance.

The project started with raw retail transaction data and involved data cleaning, preprocessing, exploratory data analysis, and data visualization using Python.

Different aspects of sales performance were analyzed, including product categories, payment methods, sales locations, popular items, and monthly and yearly revenue trends.

An interactive Power BI dashboard was also developed to present the results through KPI cards, charts, and slicers.

The project helped identify important business insights, such as the highest-revenue category, preferred payment method, best-performing sales channel, and best-selling item.

Overall, this project improved practical knowledge of Python, Pandas, data analysis, visualization, Excel, and Power BI. It also demonstrated how data-driven insights can support better business decisions.

18. PROJECT LIMITATIONS AND FUTURE SCOPE

Project Limitations

1. The analysis is based on the available retail sales dataset.
2. The dataset contains only 25 unique customers, which limits broader customer behavior analysis.
3. The 2025 data covers only January, so a full-year comparison is not possible.
4. The dataset does not provide detailed customer demographic information.

Future Scope

1. Analyze customer purchasing patterns in greater detail.
2. Include more years of retail sales data.
3. Develop sales forecasting models using Machine Learning.
4. Analyze customer retention and purchasing frequency.
5. Add more interactive features to the Power BI dashboard.
6. Provide recommendations for inventory and business planning.

This project can be expanded in the future by including more data and advanced analytical techniques.

19. PROJECT DELIVERABLES

The following deliverables were completed as part of the Retail Sales Data Analysis internship project:

1. **Python Jupyter Notebook:** Contains data loading, cleaning, exploratory data analysis, calculations, and visualizations.
2. **Cleaned Dataset:** Final cleaned retail sales data saved in CSV format.
3. **Excel Analysis Report:** Contains summary information and analysis results in multiple worksheets.
4. **PDF Summary Report:** Presents the main findings and business insights.
5. **Power BI Dashboard:** Interactive dashboard containing KPI cards, charts, and slicers.
6. **GitHub Repository:** Contains the project files, notebook, dataset, README, and requirements file.

These deliverables demonstrate the complete data analytics workflow, from raw data preparation to interactive business reporting.

20. PROJECT SCREENSHOTS

The following screenshots demonstrate the interactive Power BI dashboard developed for the Retail Sales Data Analysis project.

